MIL-B-50873A (AR)
AMENDMENT 3
31 May 1977
SUPERSEDING
AMENDMENT 2
23 December 1974

MILITARY SPECIFICATION

BAG, BARRIER, ELASTOMERIC

This Amendment forms a part of Military Specification MIL-B-50873A (PA) dated 28 July 1971 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Para 2.2; Add the following:

"D 412 - Tension Testing of Vulcanized Rubber
D 624 - Tear Resistance of Vulcanized Rubber."

PAGE 2

Para 3.1, Table I:

Delete "Tension Set 26% Max." and substitute "Tension Set 35% Max."

Delete Oil Resistance requirements.

PAGE 5

Para 4.2.1, Line 6, Change to read: "bags from each mold or mold cavity...."

PAGE 6

Para 4.2.4, Delete in its entirety and substitute the following:

"4.2.4 New equipment approval. Prior to the use of a new or reworked mold or mold cavity in regular production of barrier bags, the contractor shall submit to the Government a minimum of twenty (20) barrier bags produced from each new or reworked mold or mold cavity for inspection.

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If any barrier bag fails to comply with the applicable drawings or specification requirements, when tested as specified in 4.2.2, the affected mold or mold cavity shall not be approved for use in regular production. As a minimum, inspection shall include examination in accordance with 4.3.2.1 and drawing 9260119 and testing for moisture penetration in accordance with 4.3.3.15 (see 6.9)."

PAGE 8

Para 4.3.3.1, Delete in its entirety and substitute the following:

"4.3.3.1 Tensile strength (see 3.1), Major defect, Code No. 02001. One test specimen shall be prepared from each of 20 randomly selected barrier bags from each lot and tested in accordance with ASTM-D412 using Die C for specimen preparation. If the mean of the 20 specimens fails the applicable requirement, the lot shall be rejected."

PAGE 8

Para. 4.3.3.2, Delete in its entirety and substitute the following:

"4.3.3.2 Ultimate elongation (see 3.1), Major defect, Code No. 02002. One test specimen shall be prepared from each of 20 randomly selected barrier bags from each lot and tested in accordance with ASTM-D412 using Die C for specimen preparation. If the mean of the 20 specimens fails the applicable requirement, the lot shall be rejected."

PAGE 9

Para. 4.3.3.4, Delete in its entirety and substitute the following:

"4.3.3.4 Tear strength (see 3.1), Major defect, Code No. 04001. One test specimen shall be prepared from each of 20 randomly selected barrier bags from each lot and tested in accordance with ASTM-D624 using Die C for specimen preparation. If the mean of the 20 specimens fails the applicable requirement, the lot shall be rejected."

PAGE 9

Para. 4.3.3.5, Delete in its entirety and substitute the following:

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"4.3.3.5 Tension set (see 3.1) Major defect, Code No. 05001. One test specimen shall be prepared from each of 20 randomly selected barrier bags from each lot and tested in accordance with ASTM-D412 using Die C for specimen preparation. Elongation for this test shall be 600 percent (7" from 1" gage length). If the mean of the 20 specimen fails the applicable requirement, the lot shall be rejected."

PAGE 10

Para. 4.3.3.9, Delete in its entirety.

PAGE 10

Para. 4.3.3.11.1 with Table II, delete in its entirety and substitute the following:

"4.3.3.11.1 Ambient temperature. Barrier bags shall be randomly selected from each lot in accordance with MIL-STD-105, inspection level S-4. Each sample bag shall be tested in accordance with 4.4.1 using test equipment specified in 4.3.4. If more than one sample fails the requirements of 3.6a or 3.6b, the lot shall be rejected. If any sample fails the requirement of 3.6c, the lot shall be rejected."

PAGE 11

Para. 4.3.3.14, Delete in its entirety and substitute the following:

"4.3.3.14 Air pressure test (see 3.8), Major defect, Code No. 13001. Barrier bags shall be randomly selected from each lot in accordance with MIL-STD-105, inspection level S-4. Each sample shall be tested in accordance with 4.4.3 using test equipment specified in 4.3.4. If any sample shows evidence of air leakage through the base, sidewall or sealing rings, the lot shall be rejected."

PAGE 13

Para 4.3.5, Delete in its entirety

PAGE 15

* Para. 4.4.5, Delete in its entirety and substitute the following:

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"4.4.5 Stress relaxation. Place a 1 inch by 5 inch barrier bag specimen in a tensile tester with the jaws of the grips 1 inch apart. Impart an elongation of 200 percent to the specimen. Maintain the elongation constant throughout the test. Reduce the specimen temperature in the grips from ambient to 40 ± 2 degrees F. Allow one hour minimum for the specimen to reach the test temperature. Record the stress level. During the succeeding 40 hours minimum, continuously record the stress level. Continuously record the test temperature throughout the test. If the test temperature fluctuates more than ±2 degrees F, the test shall be considered invalid; a new specimen shall be obtained and retested. Calculate the Maximum Stress Relaxation (MSR) as follows:

$$MSR = \frac{A-B}{A}$$

where: A = Initial Stress Level

B = Lowest Stress Level Recorded

No relaxation is indicated by a MSR between 0.00 and 0.02. Repeat the entire test with a new specimen reduced from ambient to a test temperature of 20 ± 2 degrees F. Continue testing with new specimens at test temperatures of 0 ± 2 degrees F, -20 ± 2 degrees F and -40 ± 2 degrees F."

PAGE 15

Para. 4.4.7, last line; Change to read as follows:

"ATTN: DRDAR-QAT-Q (see 6.5)"

PAGE 17

Para. 6.3.4, line 15; Change to read as follows:

"..., Category A, Form 3..."

PAGE 17

Para. 6.3.5, line 9; Change to read as follows:

"DRDAR-QAT-I..."

PAGE 18

Para. 6.4, line 9; Change to read as follows:

"ATTN: DRDAR-QAM-Q"

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NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
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